TOBIAS FOLLETT

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EDUCATION

University of the Pacific Aug 2024 – May 2028
B.S., Physics GPA: 3.9/4.0 Stockton, CA

RESEARCH EXPERIENCE

Undergraduate Researcher, University of the Pacific Aug 2026 – Present
Advisor: Dr. Kieran Holland

- Implemented Monte Carlo integration and Metropolis sampling in Python/NumPy to estimate high-dimensional integrals and expectation values, validating results against closed-form solutions.
- Quantified statistical uncertainty using standard-error estimates and batch-means analysis of Markov chain output; measured acceptance rates as a function of step size.

Undergraduate Researcher, University of the Pacific Dec 2025 – June 2026
Advisor: Dr. Navdeep Singh

- Generated 1,000+ physically valid Fe–C crystal structures for MLIP training via PyXtal, enforcing volume and skewness constraints across space groups 1–230 with 2–10 atoms per unit cell.
- Executed hierarchical DFT relaxations (VOLMIN → INTMIN → FULLRELX) and systematic perturbations (hydrostatic, shear, rattle) in VASP on TAMU ACES under NSF allocation TG-MAT250003 (2.25M core-hours).
- Validated Moment Tensor Potentials in LAMMPS by computing energy-volume curves and elastic constants, comparing against reference bulk moduli for Fe_3C (202 GPa) and $Fe_{23}C_6$ (334.2 GPa).
- Developed Python pipelines for structure generation, job preparation, output parsing, and multi-stage dataset management across the HPC workflow.
- Contributed to report: "Employment of Systematically Generated Randomly Sampled Datasets to Develop Machine Learning Interatomic Potentials for Fe-C Alloys"

FELLOWSHIPS, AWARDS & HONORS

University of the Pacific AI Fellow 2026–2027

Pete and Glena Carroll Endowed Scholarship, University of the Pacific 2026–2027

Maryam H. Dashti Endowed Scholarship, University of the Pacific 2025–2026

Phi Kappa Phi Honor Society Membership 2025

Dean's Honor Roll Spring 2026

SOECS Dean's List Fall 2024, Spring 2025, Fall 2025, Spring 2026

FUNDED RESEARCH AND ACTIVITIES

NSF ACCESS Step-1 Program, Florida International University May–Jun 2026
Introductory program in cyberinfrastructure and advanced computing; 15-student cohort. Hands-on training in HPC job submission, Linux system administration, virtual machine deployment, networking, and cybersecurity.

PROJECTS

Chora 2026

- A tensor library built in the programming language Rust with basic CPU and GPU tensor operations. https://github.com/tcfollett/chora

TigerTrack: Co-Founder 2026

- Part of University of the Pacific's AI Fellows program as a university-wide opportunity to explore emerging AI initiatives at Pacific.
- Secured university funding and IT support for continued R&D and broad campus deployment.
- Architected prerequisite tracking via topological sort, a Gemini-powered AI advisor, and an automated multi-semester scheduler.
- Developed using Next.js, TypeScript, Tailwind, Firebase, and Gemini; planning integration with DegreeWorks API.

SERVICE

President, National Society of Black Engineers — University of the Pacific Nov 2025 – Present

- Lead chapter operations, meetings, and events supporting academic and professional development.

Volunteer, Math Engineering and Science Achievement Day Feb 2026

- Helped organize and lead an engineering competition for local youth.

SOFTWARE & TECHNICAL SKILLS

Simulation & Modeling VASP, LAMMPS, PyXtal, ASE, MLIP-4

Languages Python, Rust, C++, MATLAB, R

Tools & Infrastructure Git, Linux, Anaconda3, HPC (SLURM), LaTeX, Typst

COURSEWORK

Quantum Mechanics (IP), Optics (IP), Statistical Learning Methods (IP), Linear Algebra, Differential Equations, Statics, Dynamics, Mechanics of Materials, Materials Science and Engineering, Intro to Computer Science, Computer Application in Engineering, Mechanical Engineering Graphics